

Just Let Go

Non-contact inertial stabilization for the CFL space telescope

Prototype design review: architecture, actuator, sensing, control, mechanism, electronics

■ sourced ■ simulated ■ estimate / target

Bottom line

- **Architecture:** 3 independent legs. Each has a 2-axis planar Lorentz actuator (4-pole magnet + 4 PCB D-windings), payload-side accelerometers, and a bus-side laser/PSD looking at a payload cube-corner. A central controller closes 40 Hz telescope LOS.
- **Pointing is inertial, from the telescope.** The bus *follows* the payload using our relative-pose feedback (CDI / Pedreiro heritage).
- **Actuator works on paper:** 2.53 N/A per phase, 4.5 Ω , cond 1.002, 1.19 N/ $\sqrt{W}$ ($\approx 4.6\times$ CDI). $\pm 100''$ is held with mA.
- **Jitter is accelerometer-noise limited:** 282/300 nrad per axis at 0.3 $\mu\text{g}/\sqrt{\text{Hz}}$, 103/113 at 0.1 $\mu\text{g}/\sqrt{\text{Hz}}$ (target-noise suite), against CDI's 86 nrad claim. **The SI1003 reference/fallback (0.7 $\mu\text{g}/\sqrt{\text{Hz}}$) does not meet it** (simulated). Choosing the sensor is the #1 risk.
- **Proposed:** a power-loss recentering clamp. Flight heritage not established (none found in our literature set). Simulated capture in 0.34–0.38 s with no snubber hits, but only 0–2 % stress margin.
- **Not claimed:** zero-frequency isolation, whole-leg fault tolerance with 3 legs, 1 g levitation.

Requirements and what drives them

Requirement	Value	Consequence
Pointing reference	Inertial, from the telescope	Accels on the payload ; LOS outer loop
LOS feedback	~40 Hz from the telescope	Latency budget dominates stability
Angular range	±100" per tilt axis	Only ±0.14 mm at the legs, so ±0.5 mm stroke
Bus coupling	Payload → bus feedback allowed	Bus follows the payload's attitude
Unpowered behaviour	Payload pulled back automatically	Fail-safe recentering clamp
Payload	85 kg (CDI placeholder; CFL TBD)	Inertia [3.4, 3.4, 5.1] kg·m ² assumed
Deliverable	Plausible prototype by 09:00	Analytic and simulated, not built

Heritage: DFP vs CDI FFIP/DSOC

Lockheed DFP (Pedreiro 2003 → Zeledon 2026)

- Payload and bus fly in close formation with non-contact actuators
- The bus steers to follow; zero-frequency isolation *in principle*
- **Still TRL 4–5 after 20 years**; hard to ground-test

CDI FFIP → DSOC on Psyche (TRL 9)

- Collocated accel + Lorentz pairs, 80 Hz accel loop
- 4 struts (3 + 1 spare), 22.6 kg terminal
- Pyro launch locks + paraffin re-latching cages

What we take

- CDI's *inertial acceleration* loop structure (our offset sensors don't automatically inherit its collocation/positivity argument)
- Pedreiro Fig 3 topology: actuators hold translation, bus holds attitude
- DFP's kinematic ball/vee seat idea

What we fix or check

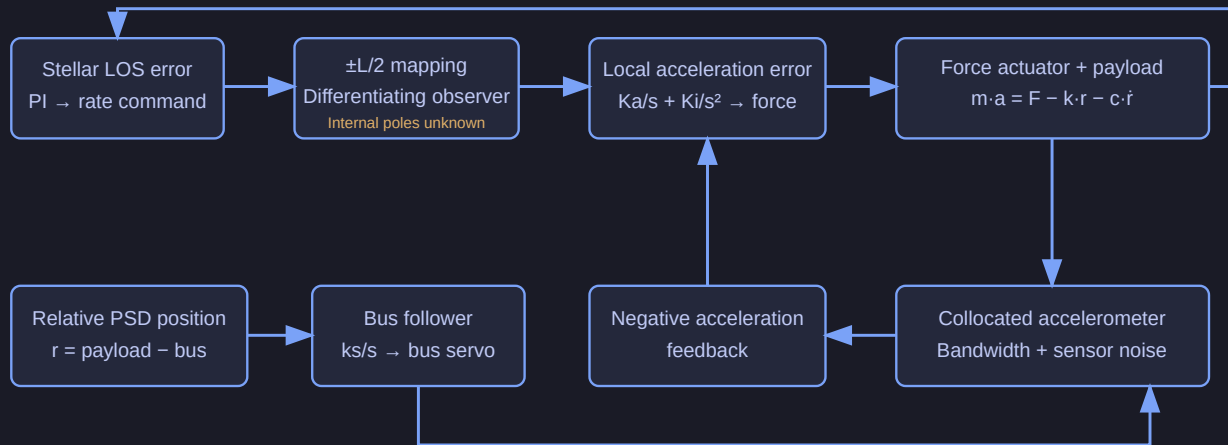
- CDI script audit: mass is 85 **kg** (unit clarification); damping $m/(\omega Q) \rightarrow Q \approx 316$ not 8; identical noise seeds
- CDI's 1 Hz umbilical = 3.4 kN/m, **~67×** our (unmeasured) 50 N/m design allocation

Sources: literature/md (Pedreiro 2003/05, Nordt 2019/20, Zeledon 2026, Andrews 2025, CDI 2012); docs/literature-digest.md; docs/control/cdi-reconstruction.md

Reverse-engineered CDI control system

CDI-derived control architecture

Original interpretation of September 2025 slides 8 and 11; not a firmware implementation



Local acceleration loops reduce inertial motion; PSD feedback moves the bus to follow.

The requested 40 Hz LOS rate replaces the deck example's 10 Hz: model latency explicitly.

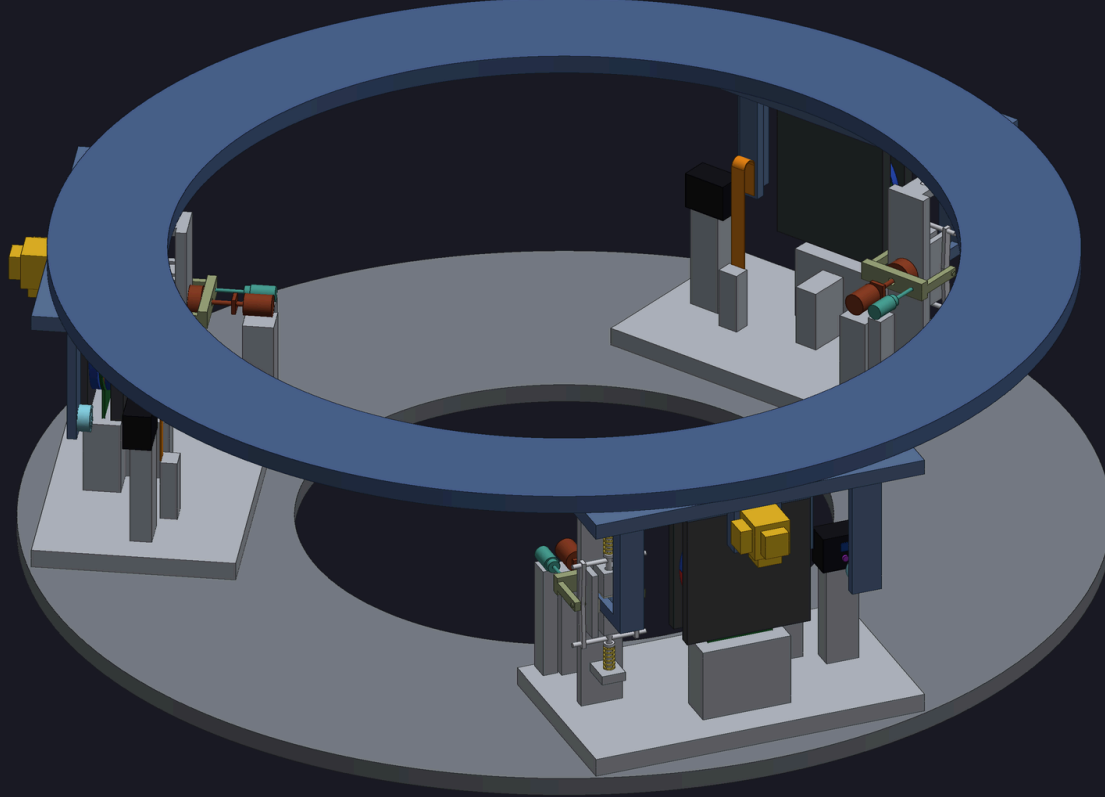
Missing source definitions: observer poles, colored-noise normalization, sampled implementation.

Diagram combines disclosed functions. Coupled rigid-body geometry and current-loop dynamics belong in the prototype model.

- Actual CDI Simulink run unmodified via **unlinked**. The accel loop is $Ka/s + Ki/s^2 \rightarrow \text{force}$ (an easy-to-miss extra 1/s). The PSD drives the **bus** to follow.
- Not supplied: observer poles and coloured-noise normalisation, so 86 nrad is **not** reproduced exactly.

System architecture

justletgo: 3 legs at 120 deg on $r_{\text{leg}} = 290$ mm (payload ring above, bus deck below), released



Three legs at 120° on $r = 290$ mm. Payload ring above, bus deck below. Each actuator plane is vertical and tangent: tangential forces $\rightarrow F_x, F_y, M_z$; axial $\rightarrow F_z, M_x, M_y$. **Rank 6 at every travel corner.**

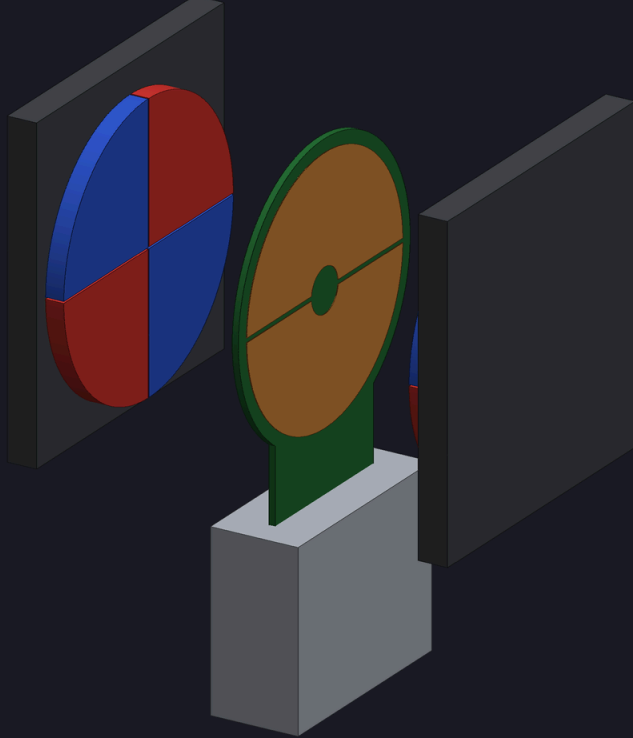
Who carries what

	Payload (floating)	Bus
Actuator	4-pole N42SH quadrant magnets, back-iron, Ti bridge	8-layer PCB coil stator
Inertial sensing	Accelerometer triad per leg (+42 mm outward offset)	—
Relative sensing	Passive cube-corner (12.7 mm)	Laser + 2D PSD (spot moves 2× payload translation, tilt-insensitive)
Electronics	Sensor electronics only (accels; SI fallback needs local conditioning)	Leg controller per leg (actuator/control power stages)
Umbilical	Flex service loop per leg	9 conductors + shield (configurable); 50 N/m design allocation ; Titan isolated reference/status commissioning open

Moving-magnet means no power crosses the gap except the accel umbilical, and coil heat stays on the bus.

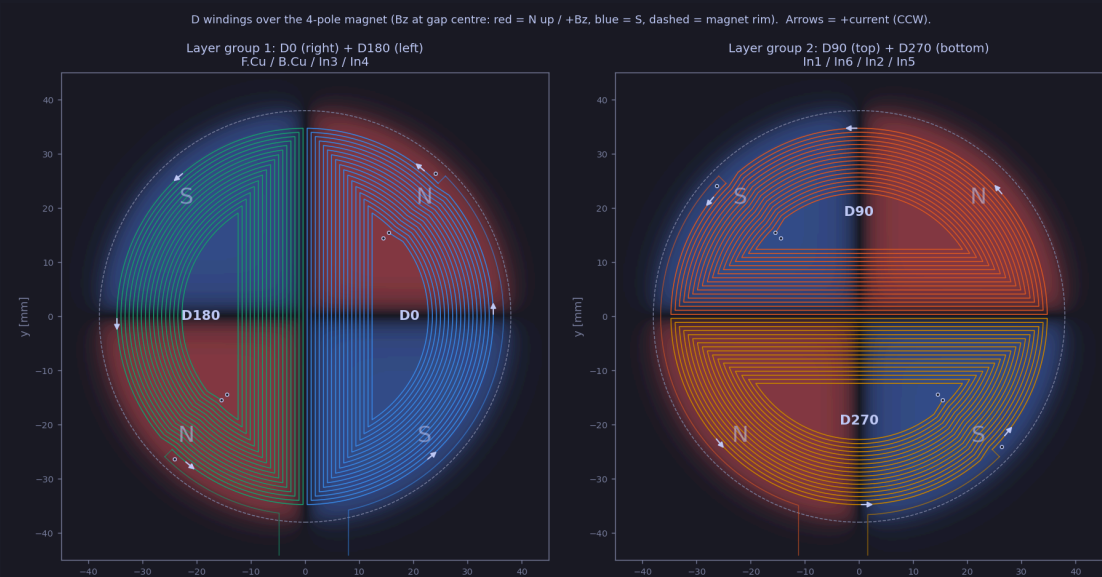
Actuator: an axial-flux motor, unrolled

Actuator exploded: $M_y = -\text{sign}(x \cdot z)$ (red +y, blue -y; same both halves) = +quadrupole in PCB frame



- **4-pole** quadrant magnet on both sides of a 4.6 mm gap. **0.87 T analytic**
- **A 2-pole magnet is rank-1**: a field varying along one axis can't push along the other. Matt's semicircle D's need 4 poles.
- 4 single-D phases at 90° on an 8-layer PCB. Differential pairs $\rightarrow F_x, F_y$; common mode $\rightarrow$ pure torque about the normal.
- ± 0.5 mm stroke; snubbers at ± 0.8 mm; coil keeps ≥ 0.38 mm clearance at a snubber corner with tilt.
- ~ 1 kN magnet attraction per leg carried by the bridge. Back-iron: **Hiperco 50, 8 mm** (≤ 2.0 T; 1018 at 10 mm for ground units).

Coil stator: 4 D-windings, 68 turns each



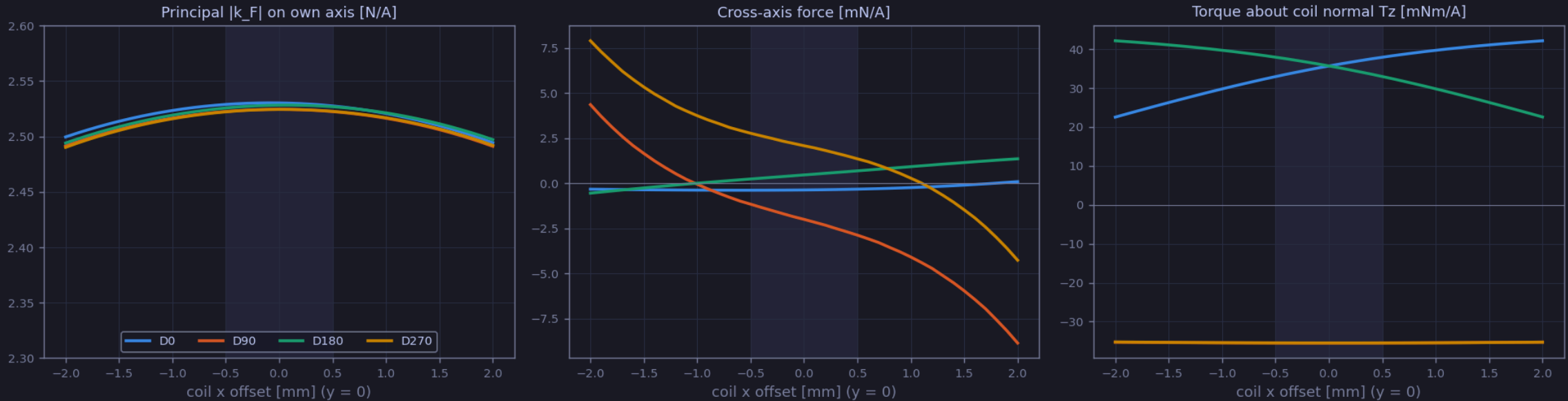
Quantity	Value
k_F per D	2.53 N/A
R per D (20/80 °C)	4.52 / 5.58 Ω
L per D	179 μH air (≤321 with iron)
Map condition	1.002
Cross residual	≤1.7 mN per N
Motor constant	1.19 N/√W (CDI 0.26)

DRC clean, same-net spacing ≥0.234 mm.

Generator: `gen_coil.py`.

Force constant over the stroke

Per-D force/torque constants vs in-plane offset, modelled 0.87 T quadrant field (shaded = ± 0.5 mm operating stroke; sweep ± 2 mm)



- Principal k_F varies **< 0.5 %** over ± 0.5 mm, so a fixed allocation suffices. The early "20 % cross term" was a midpoint-integration artifact (caught in cross-review).
- The field is **analytic** (exact quarter-disc solution, ideal iron). FEM and a gaussmeter map are needed. Back-iron flux 1.6–1.8 T $\rightarrow$ Hiperco 50 chosen.

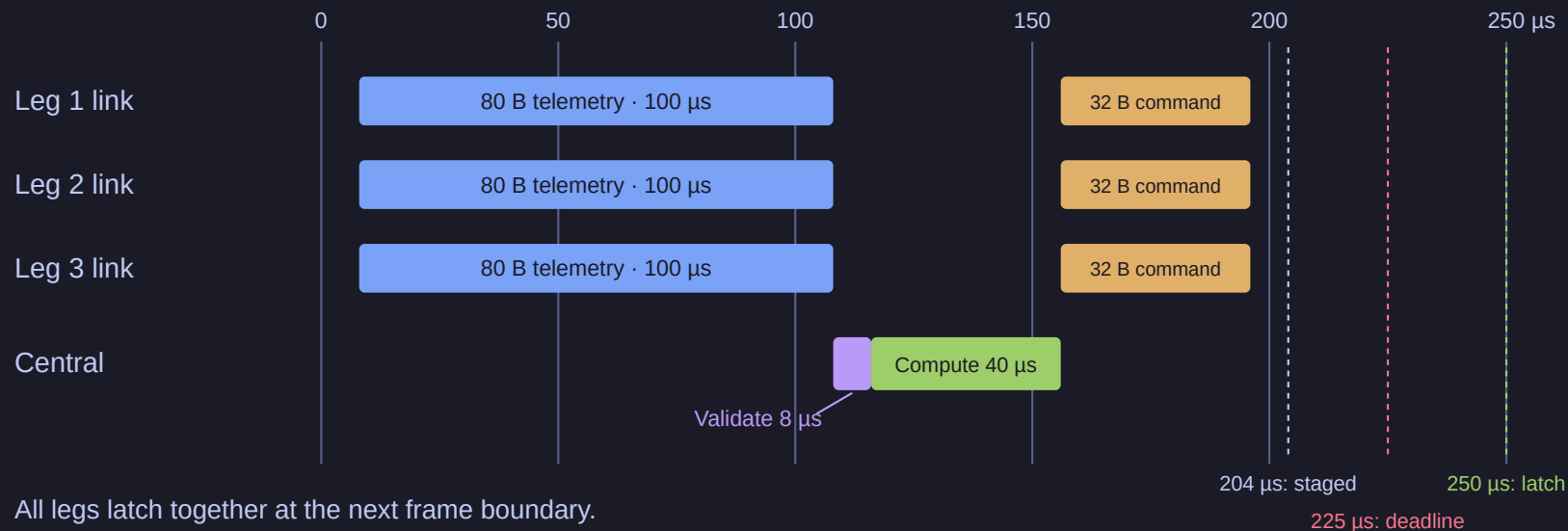
Control architecture (candidate)

- **Central at 4 kHz:** generalized inertial acceleration control (80 Hz gain, 20.1 Hz integral, CDI-style), 6-DOF observer and phase allocation. The observer blends accels (high frequency) with 40 Hz telescope LOS on the **two pointing axes** (1–2 Hz crossover); relative PSD observations complete the six coordinates. LOS integral; bias observer.
- **Local per leg:** acquire/decimate, enforce limits, apply held phase setpoints; **analog current loops are local**.
- **Relative:** PSD 6-DOF pose → **bus ACS follows** attitude (0.1–0.5 Hz, wheels ± 0.1 N·m). Translation is held by a weak ≤ 0.02 Hz PSD caging loop.
- **Fallback (bus not following):** payload-side 6-DOF PSD caging.
- **Distributed timing contract (draft):** 3 dedicated full-duplex 8 Mbps RS-485 links, one per leg. At 4 kHz each carries 80 B of telemetry and 32 B of commands (100/40 μ s serialization), with a scheduled next-frame latch. Each leg's 12 kHz DAC repeats the held 4 kHz command. Timestamps, a stale-data fault policy, and zero-on-enable.
- ⚠ Not closed: the analog current loop on the board $\neq$ the modelled PI; anti-alias and timing are unvalidated; the **central controller hardware (3 DMA UARTs + SYNC) is not yet designed** (dev board for the prototype).

Distributed control frame (proposed)

Proposed distributed-control frame

Three dedicated full-duplex 8 Mbit/s links · 4 kHz central computation · NOT measured WCET



All legs latch together at the next frame boundary.

Budget: 8 μs pack + 100 wire + 8 validate + 40 compute + 40 wire + 8 stage = 204 μs

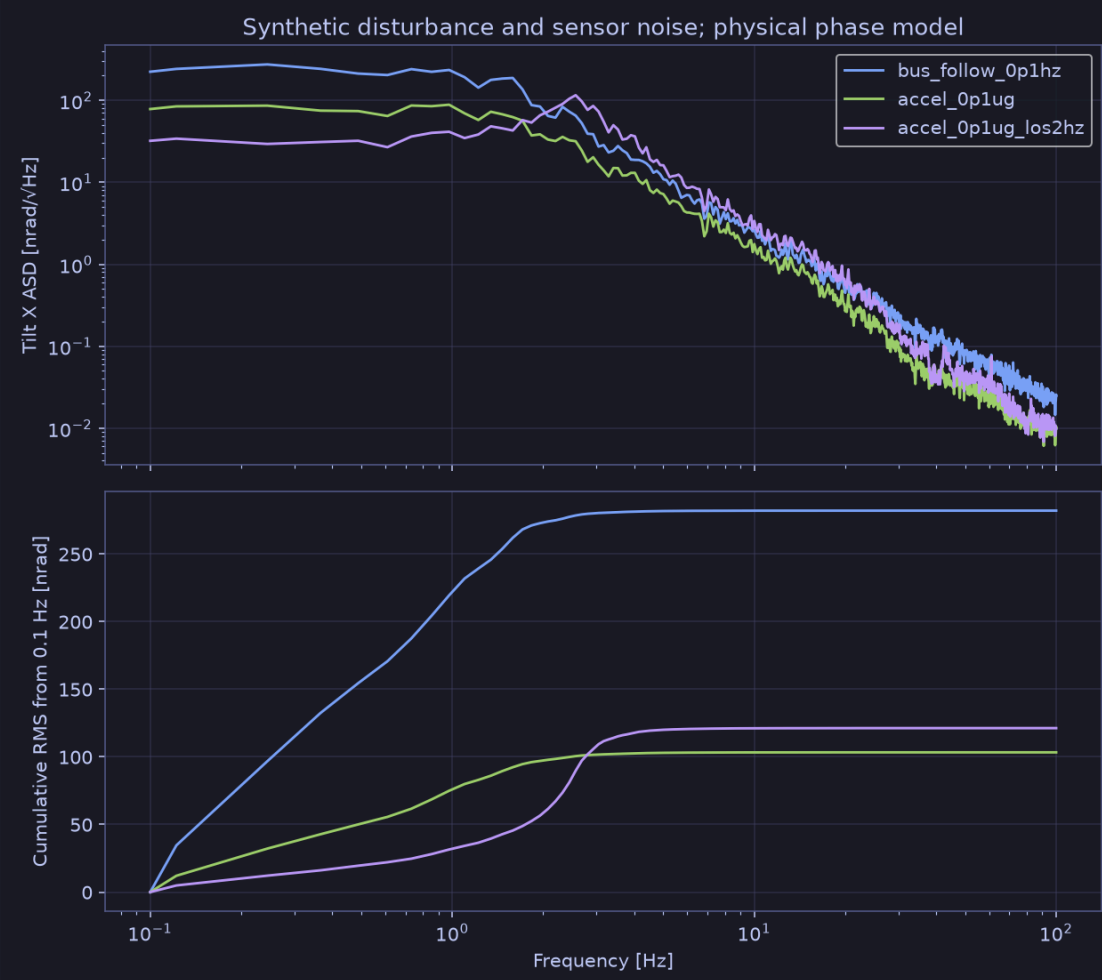
Each direction: 3.20 Mbit/s upstream / 1.28 Mbit/s downstream per leg (UART 8N1).

ADC delay 396 μs and decimator delay precede this frame; the illustrative total is 766 μs .

12 kHz local DAC writes repeat held commands. Analog current-loop dynamics still require validation.

docs/control/distributed-interface.md. Compute and serialization are budgets, not measured WCET.

LOS jitter: accelerometer noise sets the floor

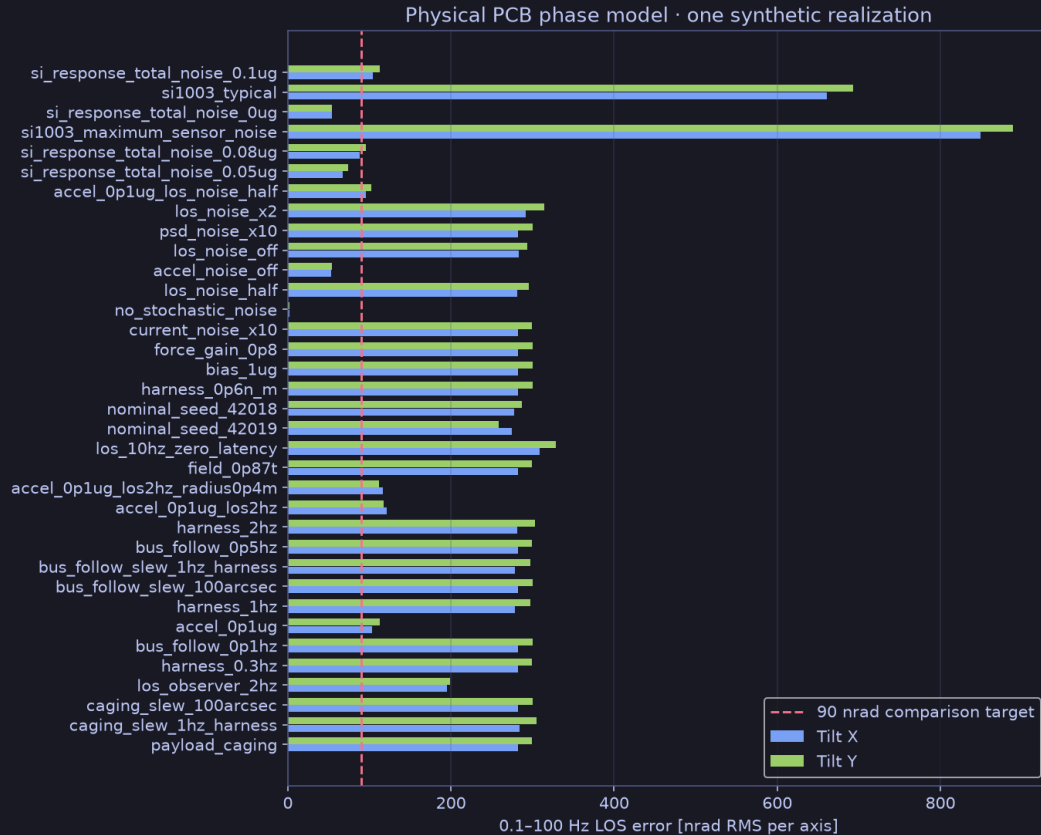


0.1–100 Hz per-axis RMS, 120 s, 12-phase model;
target-noise suite (800 Hz sensor, 812.5 μs HR ADC) (sim):

Case	X / Y (nrad)
0.3 μg/√Hz	282 / 300
0.3 μg/√Hz + 2 Hz observer	195 / 199
0.1 μg/√Hz	103 / 113
0.1 μg/√Hz + ½ LOS noise	95 / 102
accel noise off	53 / 54
no stochastic noise	1.8 / 1.7

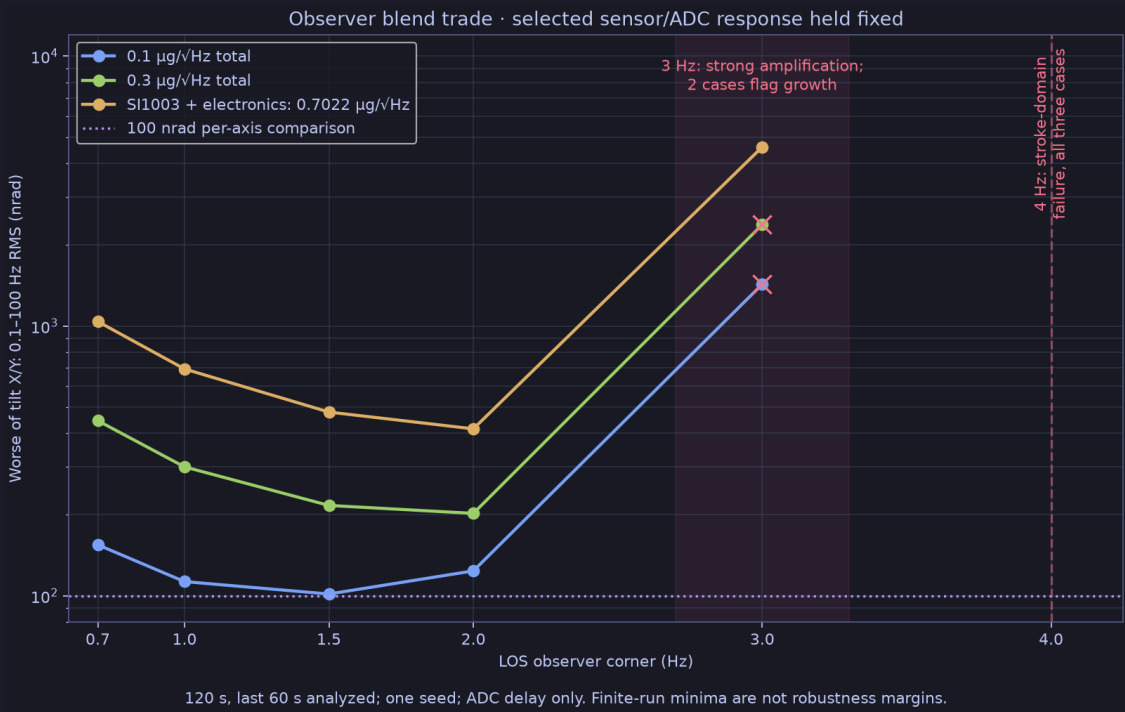
CDI claim: 86 nrad (single axis, <0.1 μg/√Hz). A separate 0.77 ms pulse test still decays at 40 Hz.

What matters (and what doesn't)



- **Dominant:** accel noise, then LOS noise and the blend crossover.
- **Negligible on jitter (tested cases only):** current noise $\times 10$, PSD noise $\times 10$, force-gain -20% , field $0.5 \rightarrow 0.87\text{ T}$, umbilical $0.3\text{--}2\text{ Hz}$ (on jitter).
- Accel lever arm $+42\text{ mm}$: $\sim 12\%$ less angular noise (rigid body only, modes not modelled).
- Blend crossover heuristic: raw double-integrated accel noise (2.46 m^{-1} lever) meets the LOS floor at $\sim 1.65\text{ Hz}$ ($0.1\text{ }\mu\text{g}$) / $\sim 2.9\text{ Hz}$ ($0.3\text{ }\mu\text{g}$). The real observer still leaks accel noise below crossover, so the **best corner is found empirically** (sweep on the next slide; Kalman/ H_2 next).

Tuning the accel/LOS blend



The LOS observer corner trades LOS noise (low f) against double-integrated accel noise (high f) (simulated, worse axis):

Accel channel	Best tested corner	Worse-axis RMS
0.1 µg/√Hz	1.5 Hz	102 nrad
0.3 µg/√Hz	2 Hz	202 nrad
SI1003 (0.70)	2 Hz	416 nrad

- **≥3 Hz amplifies strongly; 4 Hz leaves the stroke for every sensor.** Delay sets the ceiling.
- Tuning recovers ~1.6× for the SI1003, but **none of the tested tunings meets 100 nrad.**

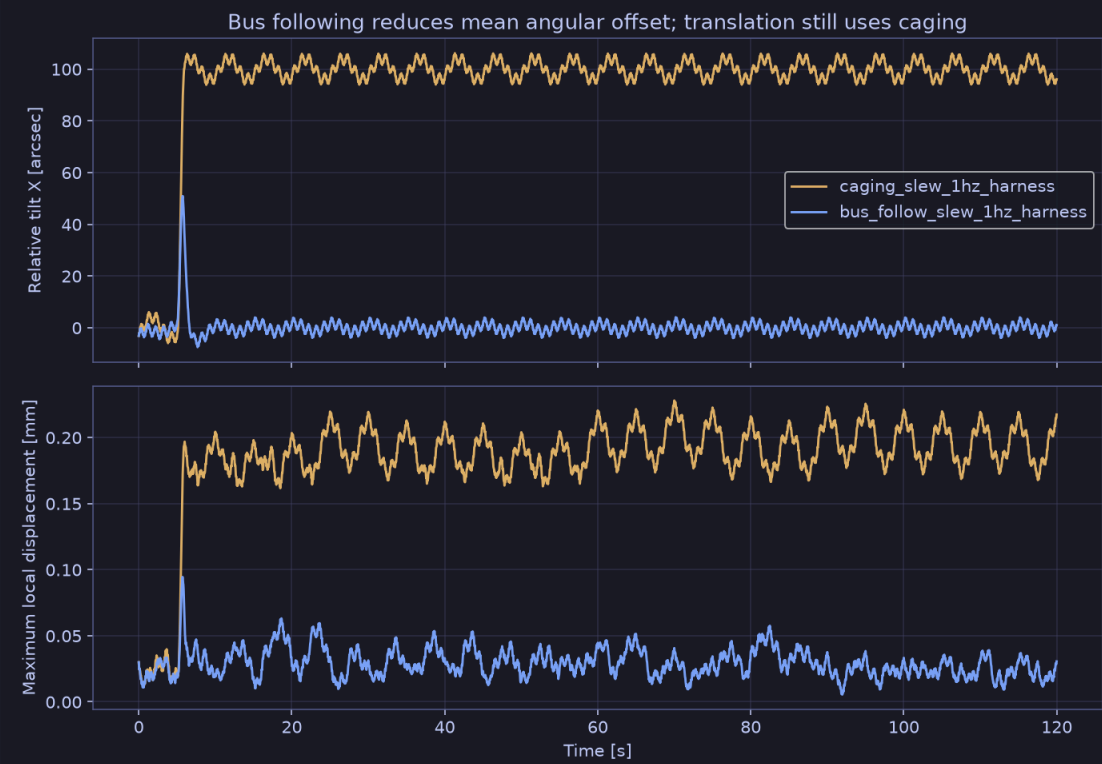
Latency, not rate, breaks the loop

Configuration	Result (simulated)
LOS 40 Hz, 25 ms latency	✓ stable, decays after pulse
LOS 10 Hz, zero latency	✓ completed 120 s run within limits (308 / 328 nrad); not a stability proof
LOS 40 Hz, 100 ms latency	✗ exceeds stroke
LOS 10 Hz, 100 ms latency	✗ unstable: growth +1.36 /s, no saturation
1 µg accel bias, no bias observer	✗ exceeds stroke

Caveat: the 4 kHz model's fractional-delay interpolation attenuates the SI1003's 1 kHz resonance by ~2.9 dB, so pulse tests validate the *modelled* discrete loop, not the real ADC/decimator resonance margin (interpolation attenuation is small near crossover, -0.017 dB at 80 Hz).

Telescope interface budget: LOS latency ≤ 25 ms (tested). Anything slower needs retuning and validation; 50 ms has not been bracketed. Failed cases are kept in the record.

Bus following: stroke and power, not noise floor

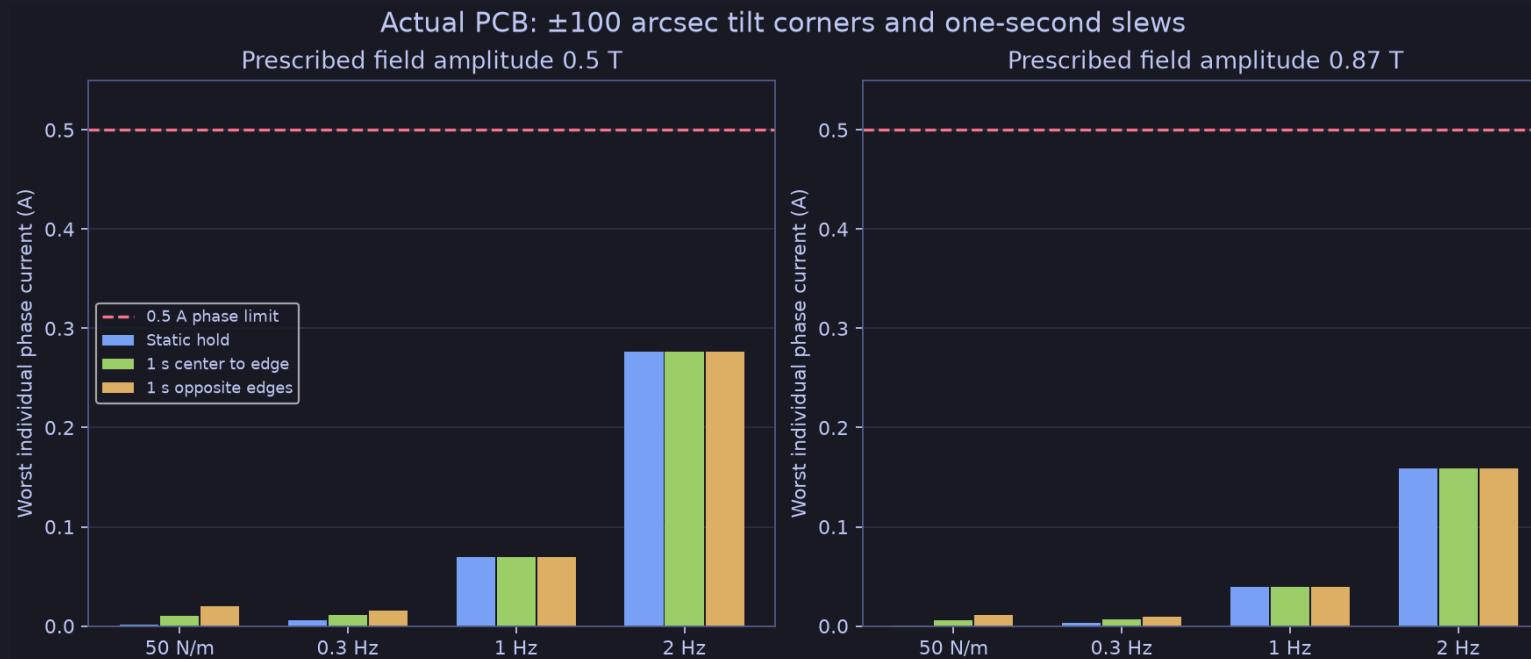


$\pm 100''$ slew, 1 Hz umbilical stress case (simulated):

	Caging	Bus follow
Peak stroke	228 μm	95 μm
Peak phase current	84 mA	48 mA
Mean copper	75 mW	7 mW

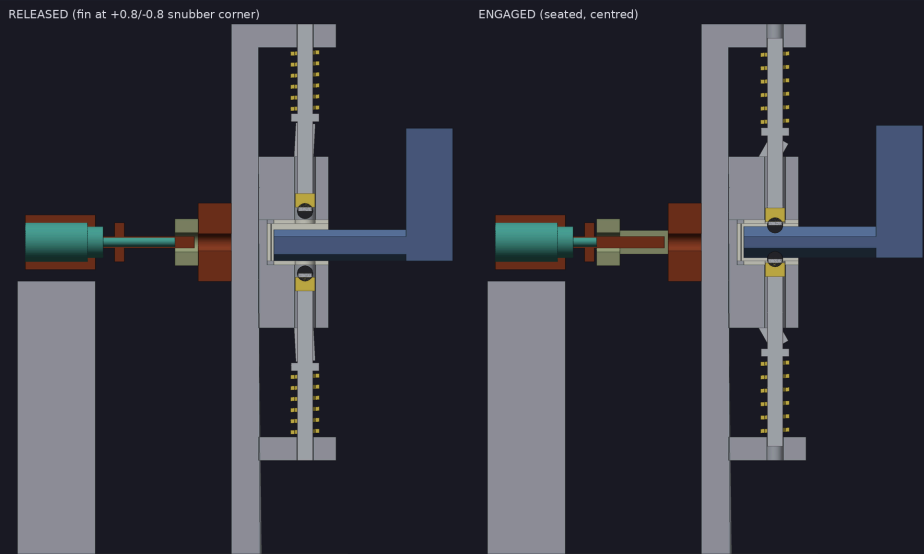
Same jitter floor (accel-limited). Wheel-activity results are withdrawn: the bus model feeds unfiltered jitter tones to the rate loop. No bus sizing claim.

Range of motion and current envelope



- $\pm 100''$ combined tilt corner = **0.192 mm** worst local motion, inside ± 0.5 mm.
- Worst hold/slew phase current: **0.159 A at 0.87 T** (0.276 A at 0.5 T), even for a stiff 2 Hz umbilical; all within the 0.5 A continuous limit.
- At the 50 N/m umbilical: **~2 mA static hold**; a 1 s $\pm 100''$ slew peaks at ~19.6 mA (0.5 T). Linear drivers **must** use low rails (a 28 V bridge would waste ~31 W).

Power-loss recentering clamp (proposed)

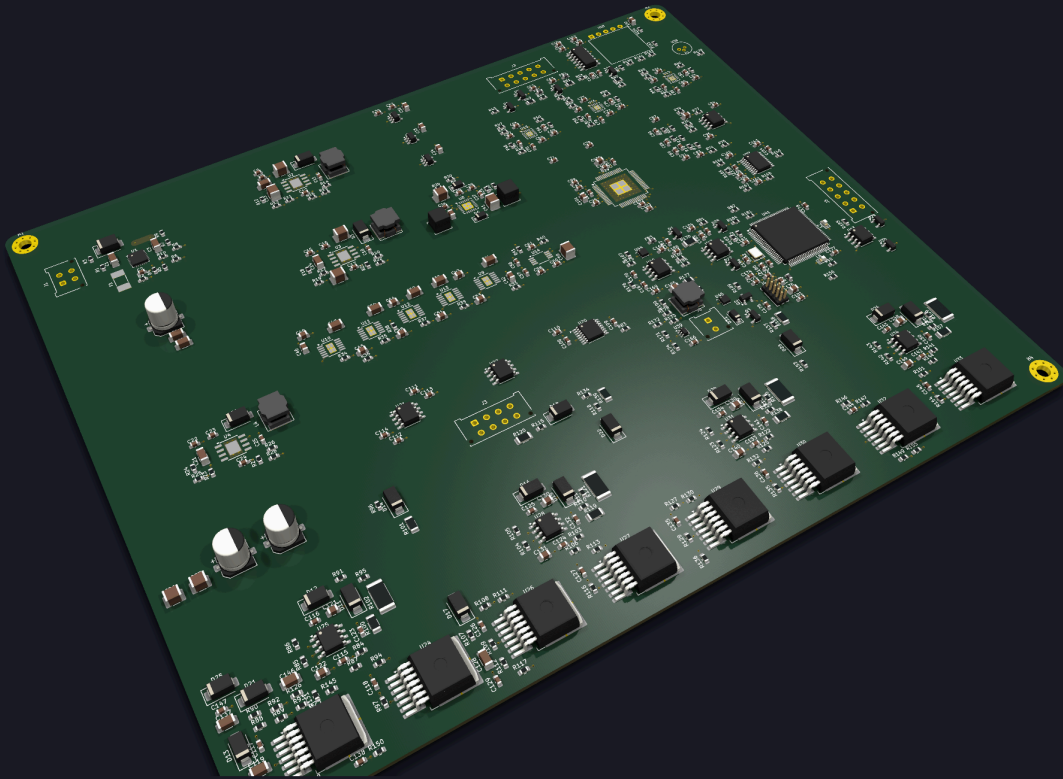


- Two spring jaws with 4 mm ceramic balls close on vees in a payload fin. A toggle is held **3° short of over-centre** by a 41 mW magnet. Powered: jaws 2 mm clear, **zero contact**.
- Power loss → the toggle collapses → a fast jaw approach, then a metering-pin dashpot near the seat. 3 sites = Maxwell kinematic seat. Not the launch lock.

Completed one-site planar cases (85 kg, 1 mm/s) (sim v1.3)	First contact	Peak stress (assumed limit 4.0 GPa)	Settled
Nominal, 7 start points	0.13–0.29 s	2.81–3.91 GPa	0.34–0.38 s
One dashpot failed	0.07–0.13 s	≤3.99 GPa	0.20–0.27 s
Both failed + crush cartridge (non-corner; corner incomplete)	0.019 s	3.3–3.9 GPa	clamped, 0.5–0.8 mm off

- No new snubber penetration (E/F/G start at the boundary); energy balance ≤0.22 %. **⚠ Open:** 0–2 % stress margin, 3-site coupling, corner double-faults, vacuum friction.

Leg controller electronics (rev A: placed, not routed)



leg controller rev A - 3D - placed + plane fanout, unrouted

- 6-layer, 200 × 165 mm, one per leg on the bus. 28 V → eFuse → bucks → LT3045/LT3094 low-noise analog rails.
- **4 BTL linear current drivers** (8× OPA548) on a 12 V drive rail, analog PI loop ~10 kHz, INA240 sense, default-off safe-enable. Worst case ≈ 7.6 W/phase vs 39 W off 28 V.
- ADS1278 HS (396 μ s) for accels + phase currents. On-board S5991 PSD, lock-in on a chopped 850 nm LD.
- STM32H743 (flight path: VA41630 / SAMRH71); 8 Mbps full-duplex RS-485 to central.
- Gates: ERC 0 · netlist 0 mismatches · DRC 0 errors · **499 connections unrouted** · quality 204 (117 intentional via-in-pad).

#1 risk: the accelerometer

Case (1 Hz observer, HS ADC, 0.87 T, real geometry)	Accel channel ASD	X / Y per axis (sim)
SI1003 reference, typical	0.7 $\mu\text{g}/\sqrt{\text{Hz}}$ $\oplus$ 0.055 electronics (RSS)	661 / 693 nrad
SI1003, max sensor ASD (typ electronics)	0.9 $\oplus$ 0.055 $\mu\text{g}/\sqrt{\text{Hz}}$	849 / 889 nrad
SI1003, best blend (2 Hz observer)	0.70 $\mu\text{g}/\sqrt{\text{Hz}}$ total	406 / 416 nrad
0.1 $\mu\text{g}/\sqrt{\text{Hz}}$, best blend (1.5 Hz)	0.1 $\mu\text{g}/\sqrt{\text{Hz}}$	98 / 102 nrad
Requirement for 100 nrad/axis, 1.5 Hz blend	$\leq 0.097 \mu\text{g}/\sqrt{\text{Hz}}$ total (sensor ≤ 0.079 assuming 0.055 electronics)	100
Requirement for 90 nrad/axis (1 Hz blend; the 1.5 Hz blend doesn't help because the LOS floor rises to ~77 nrad)	$\leq 0.072 \mu\text{g}/\sqrt{\text{Hz}}$ total	90

- We need ~7× quieter sensing than the SI1003. Datasheet-verified trade on the next slide.

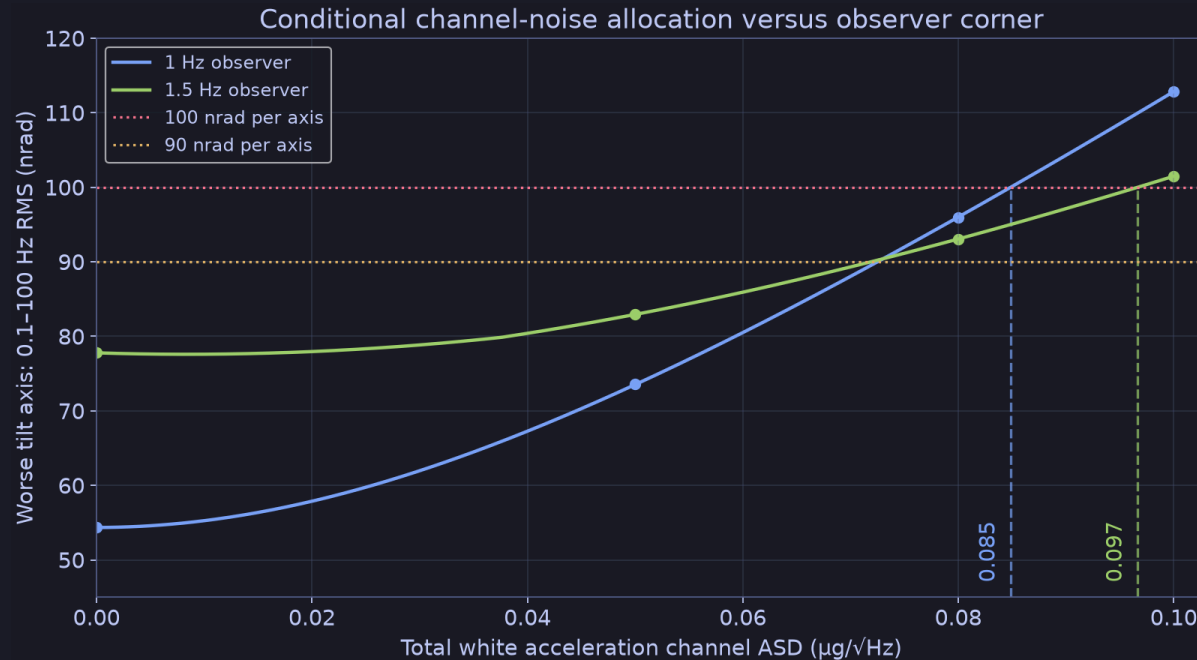
Accelerometer trade: sources and verification limits

Part	Datasheet noise ($\mu\text{g}/\sqrt{\text{Hz}}$)	Mass	Verdict
Safran SI1003 (simulated reference / fallback)	0.7 typ / 0.9 max	small	$\times$ $\sim 9\times$
Colibrys SF1500	0.3	small	$\times$, likely discontinued
Honeywell QA-3000/2000, Innalabs AI-Q	$\leq 2.2\text{--}3.2$ (upper bounds)	—	bench-test
Nanometrics Titan (triaxial FBA)	~ 0.014 (166 dB; reference unconfirmed)	0.96 kg	plausible; not qualified
Kinematics EpiSensor ES-U2	~ 0.061 (3rd-party)	—	plausible; unconfirmed

Among the candidates reviewed, no <100 g accelerometer is verified to meet $\leq 0.079 \mu\text{g}/\sqrt{\text{Hz}}$ (not an exhaustive market survey).

- **Recommended: Nanometrics Titan** force-balance: $\sim 0.014 \mu\text{g}/\sqrt{\text{Hz}}$ (166 dB, ± 4 g reference unconfirmed) $\rightarrow$ channel total **$0.021 \mu\text{g}/\sqrt{\text{Hz}}$** (conditional white-noise screening). But 0.96 kg/triad (+2.9 kg) and **no space heritage**.
- **⚠ Titan is not drop-in:** its datasheet shows $\approx -20^\circ$ phase at 80–100 Hz, right at the 80 Hz accel-loop crossover, so it needs its own transfer function and stability check. Its bias spec (± 5 mg autozero, 60/320 $\mu\text{g}/^\circ\text{C}$) means **caged calibration and thermal tracking are mandatory**.
- If the plant were otherwise unchanged (SI-shaped) at a flat $0.021 \mu\text{g}/\sqrt{\text{Hz}}$, the variance fits interpolate to **$\sim 55/58$ nrad at a 1 Hz blend** (78/77 at 1.5 Hz). **Interpolated; not a Titan prediction**.
- The front end retargets by resistor values only. Electronics noise $\approx 0.016 \mu\text{g}/\sqrt{\text{Hz}}$.

Sensor noise requirement



Same selected response, ADC delay and seed $\cdot 120/60 \text{ s} \cdot 100 \text{ nrad}$ allowance improves; 90 nrad does not.

Design target: $\leq \approx 0.097 \mu\text{g}/\sqrt{\text{Hz}}$ total accel channel $\rightarrow 100 \text{ nrad/axis}$ (1.5 Hz blend; ± 0.011 heuristic sampling scale, not a confidence interval; one seed / 60 s; ADC delay only. Scalar inner-loop screening, not a full MIMO or Titan margin: at a 0.77 ms lumped delay the phase margin drops $73^\circ \rightarrow 62^\circ$). 90 nrad needs ≤ 0.072 at the 1 Hz blend; faster blends raise the LOS-noise floor ($\sim 77 \text{ nrad}$). Conditional: one seed, white noise, ADC delay only.

Fault tolerance

- **3 legs: no whole-leg fault tolerance.** Ideal rank drops $6 \rightarrow 4$. The PCB map's "numerical rank 6" has $\text{cond} \approx 6.4 \times 10^6$, which is not redundancy.
- **Proposed 4-leg geometry at 90° (inspired by DSOC 3+1):** any single leg lost $\rightarrow$ rank 6, **cond 2.42**. Geometry only; current and transient qualification still to do.
- **Single phase lost:** still controllable at centre ($\text{cond} \approx 1.9$), but $\pm 100''$ hold needs 0.54 A at a stiff 2 Hz umbilical, over the continuous limit. Fine at 50 N/m (2 mA static hold).

Ground test plan

On the ground the 85 kg payload weighs 834 N. The clamp's assumed static seated axial capacity is 180 N (not qualified post-impact retention) and the actuators a few N, so **no 1 g levitation claim.**

1. **Bench:** coil $k_F/R/L$ vs offset (gaussmeter map), driver noise and safe-enable, accel noise PSD (validates the #1 risk)
2. **Single leg / single axis:** air bearing or long pendulum. Closed-loop accel stabilization, PSD caging, clamp capture timing
3. **Integrated 6-DOF: constant-force / counterweight or zero-length-spring offload** (<0.1 Hz needs $K < 34$ N/m; a plain spring would sag 25 m) + catch frame
4. Commissioning: a signed positive pulse per phase against the published K map (polarity is never inferred from labels)

Top risks and next steps

#	Risk	Next step
1	Accel noise: Titan is a plausible candidate; its spectrum/dynamics and space suitability are unverified	Titan measured PSD + phase → loop model; bench-test Q-Flex; space path; Kalman blend
2	LOS latency: 25 ms tested, 100 ms fails; threshold unbracketed	Telescope interface spec; retune + validate
3	Clamp: 0–2 % stress margin, 3-site coupling, vacuum friction	Coating / ball-vee resize; coupled model; tribology test
4	Field / back-iron saturation (analytic screen only)	FEM + gaussmeter
5	Umbilical stiffness unmeasured	Engberg-style stiffness test
6	Accel bias at release: a tested 3 μg single-channel residual exceeded stroke	Caged release calibration (required)
7	Distributed controller drafted, not validated; central HW not designed	Timing contract → dev-board prototype
8	No whole-leg redundancy	4th leg (proposed geometry)
9	Leg board unrouted (499)	Time-limited freerouting + 1–2 h hand route

Provenance

- Repo: `CosmicFrontierLabs/justletgo` . PRs: #1 simulation (jlg-astra), #2 literature/interfaces, #3 hardware (lab notebooks with images)
- Simulation: `sim/` , `docs/control/results/reviewed-results.json` (hashes, configs, failed cases)
- CAD: `mech/cad/justletgo_leg.dsl` (yapCAD), `mech/analysis/budget.py`
- Coil: `electronics/coil-stator/gen_coil.py` , `analyze_coil.py`
- Electronics: `electronics/leg-controller/` (kicadmium)
- CDI: hardware-documentation PR #115 (deck, `.slx` , params); reconstruction `docs/control/cdi-reconstruction.md`
- Built with the agent-portal plugins yapCAD, kicadmium, unlinked and engineering-presentations, by jlg-claude + jlg-astra with cross-review